❖ Introduction

While Image Representation involves converting an image into a suitable format or representation that can be used for further analysis or processing, Image Description refers to extracting meaningful information or features from the image for various tasks like object recognition, classification, or retrieval.

Image Description

Boundary Descriptors

such as boundary length, diameter, curvature etc.

1. Simple descriptors (length, curvature)
2. Shape numbers (the first difference of smallest magnitude)
3. Fourier descriptors (uses DFT for description)

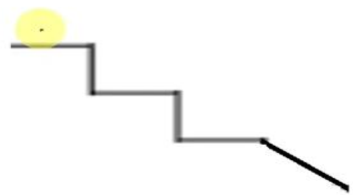
Regional Descriptors

such as area, perimeter, compactness etc.

1. Simple descriptors (Area, Perimeter, Compactness, Circularity ratio, Rectangularity etc.)
2. Topological Descriptors (measure the shape's structure without changing by deformations)
3. Texture (It provides measures of smoothness, coarseness and regularity. There are 3 approaches: Statistical , Structural and Spectral)

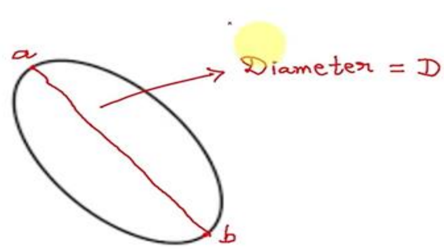
❖ (1) Simple boundary descriptors :

- **Length (perimeter):** Number of pixels along a boundary gives its length. For a chain coded curve with unit spacing in both directions, the number of verticle and horizontal components plus $\sqrt{2}$ times the number of diagonal components gives its exact length.



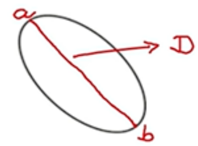
❖ (1) Simple boundary descriptors :

- **Diameter** $= \max_{i,j}\left[D(p_i, p_j)\right]$ = major axis,
  Here D is the distance measure (eucledian, cityblock or chessboard)



❖ (1) Simple boundary descriptors :

- **Major axis:** Line segment connecting the two extreme points that comprise the diameter.



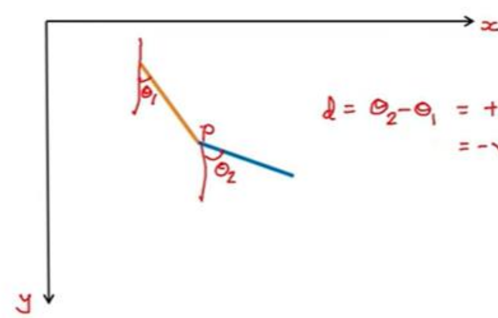
- **Minor axis:** Is the line perpendicular to major axis and 2 extreme points on the minor axis and two extreme points on the major axis should be 4 points on a box that enclose the boundary completely.

- **Eccentricity:** The ratio of major axis to the minor axis is called the eccentricity.

$$E = \frac{D}{m}$$

❖ (1) Simple boundary descriptors :

- **Curvature:** It is defined as the rate of change of slope. Curvature at the point of intersection is defined as the difference between slopes of adjacent boundary segments. To calculate the curvature, the boundary is traversed in the clockwise direction and a vertex point **P** belongs to a convex segment, if the change in slope at **P** is non-negative(i.e. +ve), otherwise **P** is said to belong to concave segment.



$d = \theta_2 - \theta_1$ = +ve $\rightarrow$ Convex segment

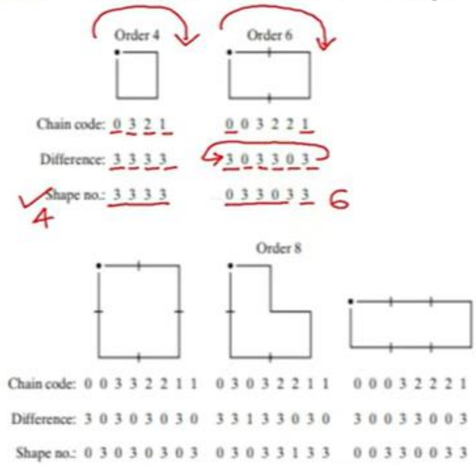
= -ve $\rightarrow$ Concave segment

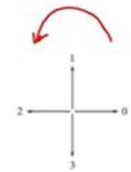
❖ (2) Shape number :

chain

- The shape number is defined as the first difference (difference code) of smallest magnitude. Order(n) of a shape number is the number of digits in the representation. n is even for a closed boundary.

FIGURE 11.17 All shapes of order 4, 6, and 8. The directions are from Fig. 11.3(a), and the dot indicates the starting point.

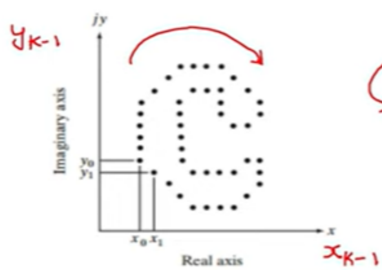




❖ (3) Fourier descriptors :

- Represent the boundary as a sequence of coordinates. They convert the object's boundary into a set of numbers in the frequency domain, capturing its essential shape features.
- Treat each coordinate pair as a complex number.

FIGURE 11.19 A digital boundary and its representation as a complex sequence. The points (x_0, y_0) and (x_1, y_1) shown are (arbitrarily) the first two points in the sequence.



2D

$$s(k) = [x(k), y(k)]\,, k = 0, 1, 2, \ldots, K-1$$

$$s(k) = x(k) + iy(k)$$

1D

- From the DFT of the complex number we get the Fourier descriptors (the complex coefficients, a(u))

$$a(u) = \sum_{k=0}^{K-1} s(k) e^{-j2\pi uk/K}, u = 0, 1, 2, \ldots, K-1$$

❖ Fourier descriptors :

- The IDFT from these coefficients restores s(k)

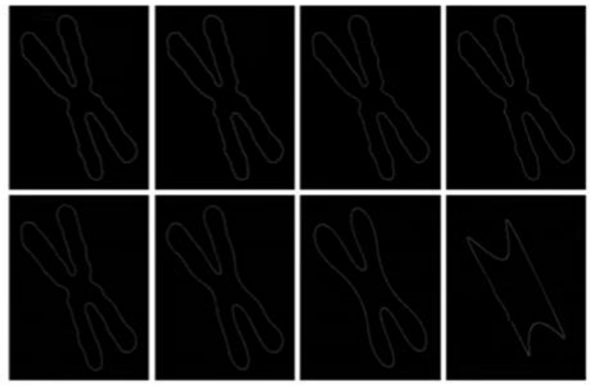
$$s(k) = \frac{1}{K}\sum_{u=0}^{K-1} a(u)e^{j2\pi uk/K}, k = 0, 1, 2, \ldots, K-1$$

- We can create an approximate reconstruction of s(k) if we use only the first P Fourier coefficients

$a(u) = 0$ if $u > P-1$

$$\hat{s}(k) = \frac{1}{P}\sum_{u=0}^{P-1} a(u)e^{j2\pi uk/K}, k = 0, 1, 2, \ldots, K-1$$

$u = 0, 1, 2, \ldots P-1$



a b c d
e f g h

FIGURE 11.20 (a) Boundary of human chromosome (2868 points). (b)–(h) Boundaries reconstructed using 1434, 286, 144, 72, 36, 18, and 8 Fourier descriptors, respectively. These numbers are approximately 50%, 10%, 5%, 2.5%, 1.25%, 0.63%, and 0.28% of 2868, respectively.

❖ Introduction

Representation is used to convert raw output obtained into another format suitable for processing the image. It deals with converting the data into a suitable form for computer processing.

Image Representation

- Chain codes
- Polygonal Approximation
- Signature
- Boundary segments
- Skeletons

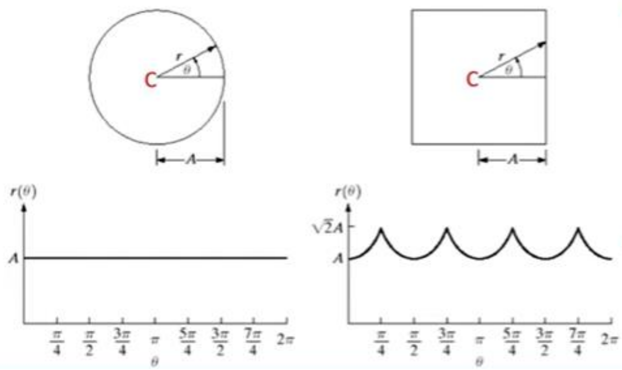
Signature: It is the sequence of normal contour sequences. It is a 1-D functional representation of a boundary.

Boundary segments: This is a type of boundary representation in which a boundary is decomposed into segments. It is used when the boundary has one or more concavities i.e, curves that carry shape information.

Skeletons: It is the representation of the structural shape of a plane region, reducing it to a graph. This reduction is obtained by skeleton of the region via a thinning algorithm.

❖ Signatures:

- It is the sequence of normal contour sequences. It is a 1-D functional representation of a 2-D boundary. It is a unique representation for different shapes and it can be used to differentiate between different objects.
- No matter how a signature is made, the main concept is to turn the 2-D boundary into a simpler 1-D function that is easier to explain than the original 2-D shape.
- The simplest signature is a plot of the distance from the centroid to the boundary as a function of angle $r(\theta)$.



a b
FIGURE 11.5 Distance-versus-angle signatures. In (a) $r(\theta)$ is constant. In (b), the signature consists of repetitions of the pattern $r(\theta) = A\sec\theta$ for $0 \le \theta \le \pi/4$ and $r(\theta) = A\csc\theta$ for $\pi/4 < \theta \le \pi/2$.

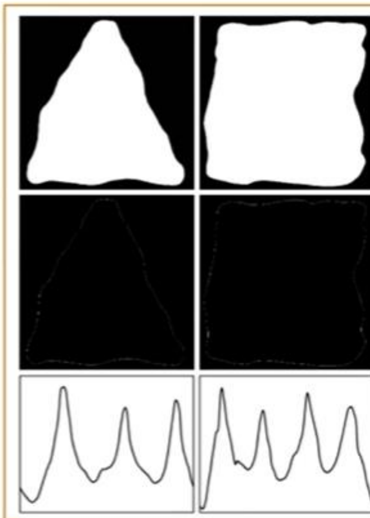
- For circle, point C is centroid and as phasor r completes 360^0, we get a one dimensional representation of $r(\theta)$ which will be constant, equal to the radius of the circle. The signature can thus be defined as 1-dimensional representation of a boundary.
- Consider another boundary in the shape of a square. In this case $r(\theta)$ will be given by the formula
  $r(\theta) = A \sec \theta$

❖ Signatures:

- Signature is an translation-invariant representation which allows an object to be compared with a standard prototype by cyclically shifting the signature of one with respect to the other in steps, while checking for the best match.
- Signatures are invariant to translation, but signatures do depend on rotation and scaling
- To make it invariant to rotation we should select the same starting point regardless of the orientation.

  – One way is to select starting point farthest from centroid (if unique).

  – Another way is to obtain the chain code of the boundary.
- To make it invariant to scaling we can normalize to a particular range.



a b
c d
e f
FIGURE 11.11 Two binary regions, their external boundaries, and their corresponding $r(\theta)$ signatures. The horizontal axes in (e) and (f) correspond to angles from 0° to 360°, in increments of 1°.